

VLSI MINI PROJECT • RTL/DV SYNOPSIS

Parameterized SPI IP Core

RTL Design, SystemVerilog Verification and FPGA Characterization

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Academic Year 2026–27 • Mini Project

SPI Connects Digital Devices

MASTER / SLAVE PROTOCOL

SPI is synchronous serial communication between a controller and one or more peripherals.

FOUR CORE SIGNALS

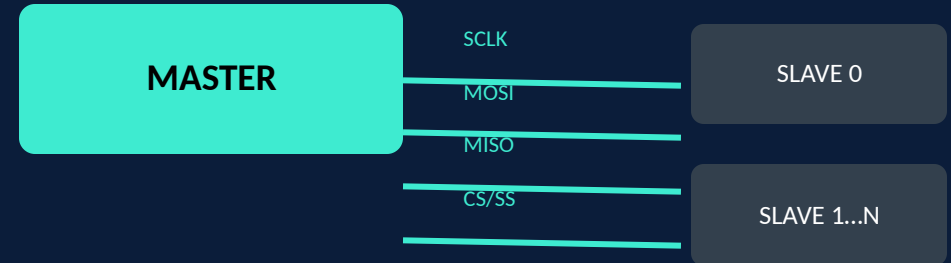
SCLK clock · MOSI Master-to-Slave · MISO Slave-to-Master · CS/SS selects the active Slave.

FULL DUPLEX, FOUR MODES

Simultaneous send/receive; CPOL and CPHA define four clocking modes.

WHY PARAMETERIZE

Data width, clock settings and slave configurations should adapt without rewriting the RTL.



CLOCK MODE STRIP

CPOL + CPHA → Mode 0 | 1 | 2 | 3

Project motivation: Combine configurable RTL, systematic SystemVerilog verification and FPGA characterization in one coherent flow.

Literature Survey

Work	Contribution	Relevance to This Project
Single Master Multiple Slave SPI on FPGA — IEEE ICAEE, 2024	Verilog multi-slave SPI with FSM control, simulation and FPGA implementation.	Baseline Master/Slave and multi-slave architecture.
Lightweight SPI Master IP for Low Cost FPGAs	Resource-conscious configurable clock, CPOL/CPHA and FPGA resource evaluation.	Motivates parameterization and hardware characterization.
SystemVerilog Verification Environment for SPI Master — ICSP, 2012	Driver, slave model, scoreboard, assertions, constrained-random tests and coverage.	Supports the structured verification methodology.
General-Purpose SPI Master/Slave IPs on OPB Bus	Configurable technology-independent Master/Slave IP and FPGA implementation.	Established configurable IP approach.
Universal Multiple Devices SPI Interface	Parameterized device selection and multi-device control.	Background for scalable slave selection.
SPI Bus on FPGA Using State Machine Method — JPCS, 2020	State-machine implementation with SPI timing-mode considerations.	Supports FSM-based RTL and FPGA implementation.
Design and Verification of SPI Bus IP Core — JPCS, 2021	IP development, structured verification and multiple timing modes.	Supports IP-oriented design and verification.

Observation: These individual techniques are established. This project integrates them into one configurable SPI IP flow and plans systematic FPGA characterization.

Problem Statement & Proposed Solution

PROBLEM

Basic SPI implementations often prioritize functional communication over reusable configuration and systematic validation.

- Different data widths, clocks, modes and slave counts
- Configuration-dependent FPGA resource and timing effects
- Manual waveform inspection alone is insufficient for systematic verification

PROPOSED

One parameterized SPI Master/Slave RTL core with 8 / 16 / 32-bit words, all CPOL/CPHA modes, configurable clock division and 1 / 2 / 4 slaves.

- Self-checking SystemVerilog: driver, monitor, scoreboard, assertions, functional coverage, constrained-random testing
- Controlled fault injection: sampling edge, bit counter and CS timing
- Planned FPGA characterization: LUTs, flip-flops, Fmax, critical path and transfer latency

Design → Verify → Implement → Measure → Analyze

Engineering Question: How do configurable parameters such as data width and slave count affect FPGA resource utilization, timing and performance, while systematic verification ensures correctness across supported SPI modes?

Methodology — System Block Diagram



PARAMETERS DATA_WIDTH: 8 / 16 / 32 · CPOL: 0 / 1 · CPHA: 0 / 1 · CLK_DIV: configurable · SLAVE_COUNT: 1 / 2 / 4

RTL BUILDING BLOCKS FSM · Shift Registers · Bit Counter · Clock Divider · Chip-Select Logic

Scope boundary: Master and Slave operate in the same FPGA clock domain. Asynchronous CDC between independently clocked domains is outside current scope.

Methodology — Transfer Flow & Verification

SPI TRANSFER FLOW

1

START /
LOAD TX DATA

2

ASSERT
SELECTED CS

3

GENERATE
SCLK

4

SHIFT +
SAMPLE DATA

5

BIT COUNTER
REACHES N?

SYSTEMVERILOG VERIFICATION

TEST / DRIVER → DUT → MONITOR → SCOREBOARD → PASS / FAIL

Directed and constrained-random tests; scoreboard checks; assertions; functional coverage.

All four CPOL/CPHA modes; 8 / 16 / 32-bit configurations; reset, back-to-back transfers, 1 / 2 / 4 slaves and data patterns.

CONTROLLED BUG DETECTION

Sampling-edge fault · Bit-counter fault · CS-timing fault

Injected Bug → Verification Failure → Waveform / Assertion Analysis → Root-Cause Identification → RTL Fix → Regression Pass

Goal: systematically verify functional correctness and demonstrate detection of realistic protocol and RTL faults.

Project Planning



Execution principles: verification begins with stable RTL blocks; feature freeze precedes FPGA characterization.

Team responsibilities: RTL Design · Verification · FPGA/Characterization · Documentation/Integration. **Target:** approximately 14 weeks.

Estimated Budget

Sl. No.	Item	Justification	Estimated Cost (INR)
1	EDA / Simulation Tools	Academic/institutional access assumed	Nil
2	FPGA Development Board	College/team-provided board assumed	Nil
3	SPI Peripheral / Components	Optional peripheral, connectors and basic hardware	500
4	Report Printing and Binding	Project documentation and submission	500
5	Presentation Material	Handouts / supporting printouts	300
6	Contingency	Miscellaneous project expenses	1,200
Total			₹2,500

Note: The estimate assumes availability of institutional EDA tools and an FPGA development board. If procurement is required, the budget will be revised accordingly.

References

1. A. H. Rahimi, A. K. Halim, A. H. A. Razak, M. F. M. Idros, F. N. Osman, S. A. M. A. Junid and S. L. M. Hassan, “Design and Analysis of Single Master Multiple Slave Serial Peripheral Interface (SPI) on FPGA,” IEEE ICAEE, 2024, DOI: 10.1109/ICAEE62924.2024.10667616.
2. M. B. Aykenar, G. Soysal and M. Efe, “Design and Implementation of a Lightweight SPI Master IP for Low Cost FPGAs.”
3. Z. Zhou, Z. Xie, X. Wang and T. Wang, “Development of Verification Environment for SPI Master Interface Using SystemVerilog,” ICSP2012 Proceedings, 2012, DOI: 10.1109/ICoSP.2012.6492015.
4. “Design and Test of General-Purpose SPI Master/Slave IPs on OPB Bus.”
5. T. Liu and Y. Wang, “IP Design of Universal Multiple Devices SPI Interface.”
6. J. Qiang, Y. Gu and G. Chen, “FPGA Implementation of SPI Bus Communication Based on State Machine Method,” Journal of Physics: Conference Series, vol. 1449, 012027, 2020, DOI: 10.1088/1742-6596/1449/1/012027.
7. D. Wang, J. Yan and Y. Qiao, “Design and Verification of SPI Bus IP Core,” Journal of Physics: Conference Series, vol. 1971, 012032, 2021, DOI: 10.1088/1742-6596/1971/1/012032.

THANK YOU

Questions & Discussion

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MASTER — SCLK — MOSI — MISO — CS — **SLAVE**